

Compute Ledger - GPU Allocation Policy

System assumptions

$$20 \text{ GPUs} \times 24 \text{ h/day} \times 90 \text{ days} = 43,200 \text{ GPU-hours}$$

- This is a ceiling, i.e, planning assumes well below 100% utilization; idle time, maintenance, and queueing are normal.
- **Modeled on Slurm-style clusters:** researchers submit batch jobs, monitor status, cancel as needed. Quotas, priority, preemption, and reserved capacity are all enforced server-side, visible only as the limits and notifications a researcher experiences.
- Researchers register once with the resource allocation system and receive cluster credentials, head-node access and per-user storage for checkpoints and outputs.

Jobs are categorized based on duration as follows:

Title	Duration	Typical Use
Small	0 - 36 hours	Short experiments and/or Evaluation jobs
Medium	36 - 72 hours	Finetuning, mid-length experiments
Large	3 - 7 days	Long unattended batch runs

1. Request and approval workflow

- Once registered, a request to submit a job may contain the user ID, number of GPUs, maximum duration of the job.
- **Small/medium jobs ($\leq 72\text{h}$):** auto-admitted the moment GPUs and quota (§2) are available — no human review.
- **Large jobs ($> 72\text{h}$):** require PI sign-off from the GPU admin before submission is accepted, keeping the highest-impact jobs under review while routine work flows through unattended.
- Auto-admitted jobs start within the scheduler's normal loop (under a minute once free); large-job sign-off targets same-business-day turnaround.

2. Fair-use constraints

- Fairness is enforced on **cumulative GPU-hours per PI over the full grant**, not per-job limits as a per-job cap alone doesn't stop continuous resubmission over 90 days.
- 15% of budget (6,480 hrs) is held as a flexible reserve; 85% (36,720 hrs) is split as baseline per-PI quotas (~ 367 hrs/PI), adjusted by declared usage pattern (eval-heavy vs. pretraining-heavy).

- Submissions exceeding quota are rejected outright at request time.
- Unspent quota returns to the reserve, opened for reallocation in the grant's final month.
- Admins retain a logged manual override for exceptional cases.
- **Per-job guardrails:** defaults to max 2 GPUs and 24hr duration; larger requests need to go through the large-tier path.
- **Queue priority:** combines age-in-queue with fairshare (recent heavy usage lowers priority, recovering as usage tapers). This also naturally favors requesters with fewer active jobs, without a separate rule.

3. Preemption rules

- **Self-overflow:** Jobs are automatically terminated at the wall-clock limit, with warnings at 50% and 80%; admins may grant up to a 6-hour extension if capacity allows.
- **External preemption:** Higher-priority jobs trigger a short warning for checkpointing, after which the job may be force-stopped and queued to resume when capacity is available.
- **Preemption-safety:** long-running jobs must implement checkpoint/resume and handle SIGTERM gracefully, writing to a fixed path convention (`<user_id>/<job_id>/checkpoint/`) so the system and the admin can verify a checkpoint exists.
- **No refund for lost work:** GPU-hours are charged for time held, regardless of checkpoint success — mirrors real billing (occupancy, not completed work) and incentivizes checkpointing without policing code.

4. Budget tracking

Tracked in GPU-hours per job, aggregated per PI and system-wide.

- **Burn rate:** Calculated as rolling 7-day average vs assumed baseline ($43,200 \div 90 \approx 480$ hrs/day). Values $>1.5\times$ baseline triggers an early warning, and values $<0.5\times$ flags underused capacity for possible reallocation.
- **Monthly pacing cap:** any month exceeding 15,000hrs (vs assumed $\sim 14,400$ hrs monthly target) alerts independently, catching a fast month early.
- **Absolute threshold:** at 80% consumed budget, all users and admin are notified, priority shifts toward PIs with more quota left, and have more jobs to run. At 95–100%, new large-tier approvals freeze except by override.
- GPU hardware telemetry and job/budget accounting can feed one dashboard so that hardware and budget problems are both visible from one place.

5. AISEhack burst (72h hackathon, 200 external participants)

- **Assumption** - This is tracked on a **separate account/budget** and doesn't draw against the 100 researchers' 43,200hr allocation.
- **Capacity carve-out:** admin reserves GPUs exclusively for the window, ahead of time.
 - 15 of 20 GPUs reserved for the hackathon.

- 2 GPUs may be reserved for up to 2 exclusive high-priority research jobs (1 GPU each), unaffected.
 - 3 GPUs held flexibly, assigned to whichever pool (research or hackathon) shows consistently higher demand after day 1–2.
- Researchers will be notified 7–10 days ahead, and confirmed 2 days prior to the event. Participants offload non-GPU work to CPU and free sessions promptly.
- **Access model:** $15 \times 72\text{hrs} = 1,080$ GPU-hours across 200 participants (~5.4hrs/person if split evenly) simultaneous access for all 200 isn't physical on unshared A100s.
 - GPUs may be served via **time-boxed rotating slots** (45-min default runtime); GPU returns to rotation the moment a slot ends.
 - Heavier users naturally queue behind lighter ones.
 - MIG partitioning (where supported) can further increase concurrency.
 - Organizers may permit external resources (e.g. Colab) alongside reserved hardware.